

The Precise 3D Reconstruction of Human Faces Based on 2D Photograph

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Abstract—3D object reconstruction is a broad research area in the computer vision field including complicated image processing techniques and existing open problems. One of the existing challenges in this field lies with accuracy, thus most works are designed for architecture reconstruction where accuracy is not the key. The purpose of this work is to discuss a flexible scan system using multiple cameras that can correctly reconstruct 3D objects - a human face with expression. Further, a mathematical model is proposed in this literature to automate the 3D object reconstruction process.

Keywords—photogrammetry, 3D reconstruction, Camera, Multiple Views, 2D images, depth perception

I. INTRODUCTION

3D reconstruction using photogrammetry techniques, which uses methodologies to capture and reproduce the shape and appearance of an arbitrary object or scene given depth and color information, is a broad research area within the computer vision field involving many open problems. Photogrammetry is the opposite of photographing, with depth data added to the 2D images. Other words, it reconstructs a 3D object from a set of 2D photographs or videos. The appearance of a three-dimensional (3D) object is determined by the shape as well as texture of the object. Certain obstacles that an object reconstruction system must overcome are different in results due to variations in illumination, viewing angle, micro movement, occlusion [1, 2], and changes over time.

This proposed research will benefit many practical areas, including but not limited to biometric identification, authentications, cyber security, preservation of historic artifacts, drama acting especially those with rapid and complex facial gestures, and many others. The rest of the paper is organized as follows:

1. Section 2 provides a brief survey of comparable techniques existing in this field.
2. Section 3 discusses our methodologies and algorithms for effectively reconstruction of 3D objects. It also discusses the implementation and the testing of the developed methodologies.
3. Section 4 and 5 concludes with lessons learned and some final thoughts.

II. BACKGROUND

A. Similar Tools

Similar work can be found on various platforms, and they cost differently. Most of the tools listed here are free, or some provides a free trial version:

TABLE I. EXISTING TOOLS AND THEIR COMPARISONS

Name	Type		
	Important Information	OS	Price
PhotoCatch	Transform video and photos into 3D models.	iOS, macOS	Free, but 9.99 for pro
KIRI Engine	Designed for phone, can connect to 3D printer	iOS, Android, Browser	Free (3 downloads per week)
COLMAP	offers general-purpose Structure-from-Motion and Multi-View Stereo pipeline with a graphical and command-line interface.	Windows, macOS, Linux	Free
Meshroom	Aerial, Close-Range	Windows, Linux	Free, based on AliceVision framework
MicMac	Mainly for professional or academic users	Windows, macOS, Linux	Free
OpenMVG	Developed in C++, it is a library for computer-vision and Multiple View Geometry projects	Windows, macOS, Linux	Free
Regard3D	Converts photos of an object, taken from different angles, into a 3D model of this object.	Windows, macOS, Linux	Free
VisualSFM	3D reconstruction using structure from motion (SFM)	Windows, macOS, Linux	Free
3DF Zephyr	reconstruct 3D models from photos	Windows	Lite version \$149, 500 photos limit
Autodesk ReCap	Infrastructure and building design	Windows	\$3270/ year
Trnio	Face scanning possible.	iOS	Free from the app store
Scann3D	Peroforms photogrammetry locally	Android	€40.99/year

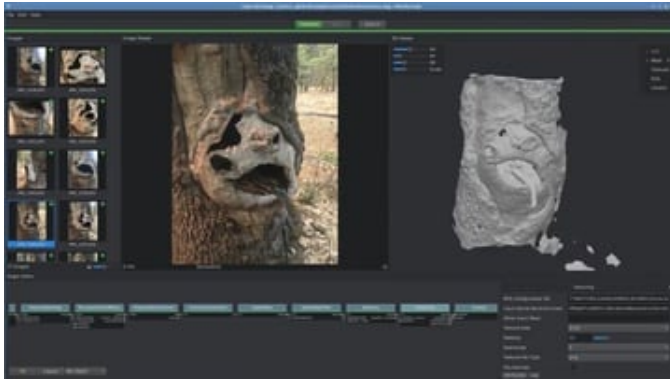


Fig. 1. MESHROOM (<https://alicevision.org/>)

B. Challenges

Despite the early initiative works in 3D object recognition in the late 1980s [3, 4, 5], relatively few researches have focused on this area during until the early 2000. In the last a few years, interest in 3D Object reconstruction research using automatic or semi-automatic techniques was resurrected [6, 7] and has increased rapidly since then, largely due to the evolution of machine learning techniques [8] and the advancement of big data. The increased demand of high-quality 3D object reconstructed models of complex physical objects grows in a wide range of applications such as gaming, medical, entertainment, cultural, architectural [9]. A flexible scan system using multi cameras that can correctly reconstruct 3D stable or moving objects like drama acting and dancing, with facial expression of actors is useful. For example, the 3D reconstruction of faces can be copied to animation performers that will dramatically change the anime creation.

Another reason is the widely adopted and affordable hardware, such as smart phones and digital cameras, which have made the acquisition of high-quality and high-resolution images much more ubiquitous. Despite this growing attention, the current state-of-the-art 3D object recognition systems perform well when objects are captured as a 2D image. However, the development of 3D object reconstruction systems that can precisely reconstruct a human face is still an open research issue.

One of the existing challenges in the field of human face reconstruction lies with accuracy, such as micro motion, or the facial expressions on the appearance of the human or animal face, which has greater details compare to object such as a building. Similar literatures in this field focuses on 3D reconstruction in stable conditions such as an existing image or photos taken in a rather static environment.

More specifically, the main challenges of human faces reconstruction based on 2D images are:

- 2D image lost the information of rise and fall of face along project direction.
- The image is sensitivity to the direction and intensity of environment lights.
- The same people may have different facial expressions.
- The same face may show various images based on the positions of camera during photograph.

These challenges can be avoided or much reduced by taking 3D information of a face. Object recognition research using automatic or semi-automatic techniques received significant attention in the last two decades. This purpose of this research project is to conduct a flexible scan system using two cost effective cameras that accurately reconstruct 3D stable objects.

III. METHODOLOGY

This section describes basic methodologies for this reconstruction mechanism. To start with, we have two cameras from two different angles. This work provides an open-source free application.

A. Principle

To scan or measure the 3D data of an object using 2D film of camera, the spatial relationship between camera and object must be known. Figure 2 shows the geometrical positions of object and camera, where (X, Y, Z) on the object is the absolute coordination system that the object is located. (x_{s1}, y_{s1}, z_{s1}) represents the spatial position of focal point S of camera in the system. f on the image represents the perpendicular distance between the focal point and the 2D image. (x, y) is the 2D coordinate system of 2D image. Simple object with uneven surface, a mannequin in this case, is used to represent a human head (Figure 2). In figure 2, o , which is shown on the 2D image, represents the perpendicular projection the focal point of the camera.

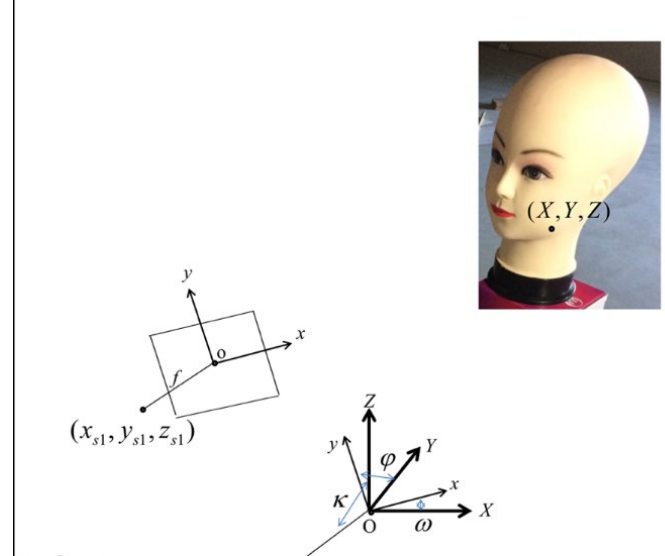


Fig. 2. The mannequin shown in this figure represents a potential human object (or live object) to be captured reconstructed and hence recognized

If camera system is nonparallel to absolute system, 3 angle elements (ω, ϕ, κ) as Figure 2 shows are needed, where the original point of camera system was moved to the original point of absolute system.

Every point on the object $(X1, Y1, Z1), (X2, Y2, Z2), \text{ etc.}$ produces a project point $(x1, y1), (x2, y2), \dots$ on 2D image as Figure 2 shows. Both groups of point have a fixed relationship determined by elements $(x_{s1}, y_{s1}, z_{s1}, \omega, \phi, \kappa, f)$. The relationship is as following:

$$(x_i, y_i) = F(x_{s1}, y_{s1}, z_{s1}, \omega, \varphi, \kappa, f, X_i, Y_i, Z_i) \quad i = 1 \sim n \quad (1)$$

Where n is the number of points

Inversely, if the points of object (X_i, Y_i, Z_i) are known and the points in film (x_i, y_i) can be measured, other 7 elements can be expressed as:

$$(x_{s1}, y_{s1}, z_{s1}, \omega, \varphi, \kappa, f) = C(x_i, y_i, X_i, Y_i, Z_i) \quad (2)$$

Using the two equations provided above, if n is enough ($n > 7$), we can use non-linear least squares to solve the equation above. When the 7 elements are known, the inverse function of function (1) cannot be used to calculate (X, Y, Z) that includes 3 unknown elements from (x, y) that has only 2 known elements as shown in equation 3.

$$(X_i, Y_i, Z_i) = G(x_{s1}, y_{s1}, z_{s1}, \omega, \varphi, \kappa, f, x_i, y_i) \quad (3)$$

Therefore, to obtain (X_i, Y_i, Z_i) , there must be more than one camera. Figure 3 shows the 3D scan system using two cameras. The relation between each point of the 3D object and the two cameras can be explained as:

$$\begin{bmatrix} X_1 \\ Y_1 \\ Z_1 \end{bmatrix} = f \begin{bmatrix} x_{11} & y_{11} \\ x_{21} & y_{21} \end{bmatrix}_{x_{s1}, y_{s1}, z_{s1}, \omega_1, \phi_1, \kappa_1, f_1, x_{s2}, y_{s2}, z_{s2}, \omega_2, \phi_2, \kappa_2, f_2} \quad (4)$$

Where x_{11} and y_{11} represent the first camera and x_{21} and y_{21} represent the second camera. (x_{11}, y_{11}) and (x_{21}, y_{21}) should represent the same point on the 3D object on two different 2D images taken by two different cameras.

This mathematical model serves as the foundation of this project.

B. Preliminary Test

Computer graphics can produce synthetic 3D models for some of these applications by a combination of artistic and technical design processes. Conceptually, obtaining a 3D object reconstruction in a stable condition (e.g. rooms with constant lighting, non-movable objects) is easier. To better address the problem, we started with a set of preliminary experiments to build and discover a relationship between camera position, objects and their relationships in an absolute 3D space. The experiments are carried out so a preliminary mathematics model described in the previous section.

Equipment and supplies used in this preliminary test: two cameras (\$40 each) and random objects at home with an uneven surface (a mannequin). The cameras are set up in a room with natural lighting. As shown in table 1, cameras are widely used in computer vision field for 3D object reconstructions. Numerous applications require a similar computer vision approach to object surface reconstruction.

To know (X_i, Y_i, Z_i) , we can measure some marked points in space, or we can create some points using a special device such as a calibrator, as shown in figure 3. The edges, corners and points on the calibrator can be measured before the testing.



Fig. 3. Finding the location of the camera



Fig. 4. The angles of each camera are recorded, Notice this is the angle in the absolute 3D space

As shown in figure 4, we utilized 2 cameras in this experiment. The cameras' locations are calculated based on equation 4. Typical mobile or desktop devices with a build-in camera can also be used in this application. The benefit of using two camera is to capture two images simultaneously, so micro facial expression differences due to time delay can be neglected.

One of the major early challenges we face was the number of cameras. Simple object with uneven surface, a mannequin to represent a human head in this case (Figure 5), cannot be reconstructed with full details by just using one simple camera as described in the previous section. Most of the applications described in table 1 uses multiple 2-D photos taken by a single camera. Some other methods require a laser pattern projector, an IR camera, and a color, such as those used in a Kinect System [10], which can cost hundreds if not thousands of dollars and they are rather high maintenance. To better capture the image three dimensionally, we first projected an image with a set of simple 2D lines onto the object and quickly obtained a set of uneven curvy lines, which are used for identify correlated (X_i, Y_i, Z_i) in a closer range. This 2D lines are used for speeding up the process in finding (X_i, Y_i, Z_i) .

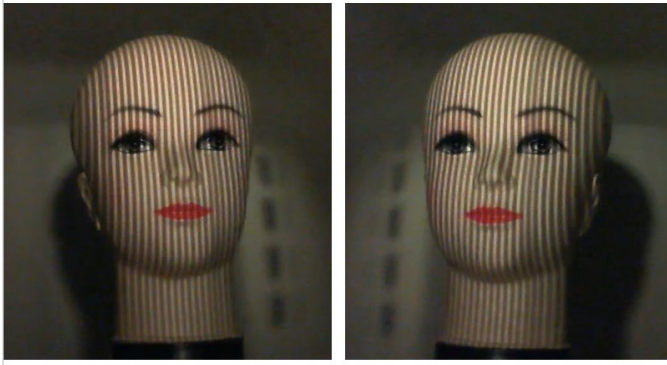


Fig. 5. The angles of each camera are recorded, Notice this is an angle in the absolute 3D space

Another difficulty during this process is how to identify the same point (X_i, Y_i, Z_i) on two different images when the locations of the two cameras are known. The 2D lines are used to assist finding the same point on the object from two different angles using Digital Image Correlation [11]. The lines and their shapes will help to identify object's 3D construction in pixels. The lines are necessary for narrowing the range when identifying correlated points on the two images taken from two different angles. Notice prior literature supports a simple object with completely even surface can be easily reconstructed [12].

C. Scalability Testing

We tested the proposed methodology using a human subject with micro facial expressions but no occlusion. with the two recorded angles and their distance from the camera, the resulting coordinate of each projected line in an absolute 3D space can be obtained, as shown in Figure 6.

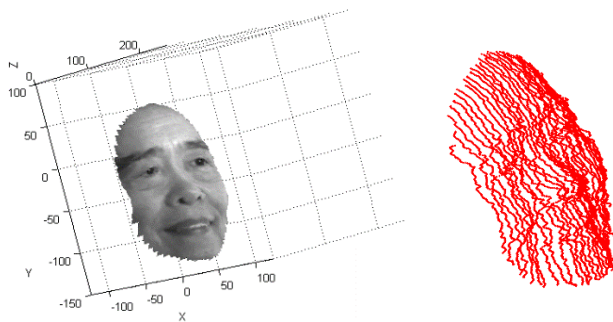


Fig. 6. 3D projected result and the lines

In figure 6, the red lines represent the projected lines on the human face. As described in the previous section, the lines are needed so we can find the corresponding pixels, or the (X_i, Y_i, Z_i) in a closer range. The blue dots in figure 8 represents all (X_i, Y_i, Z_i) identified in the Epipolar plane. Each pixel on a figure has a location (X_i, Y_i, Z_i) and the corresponding color (R, G, B)

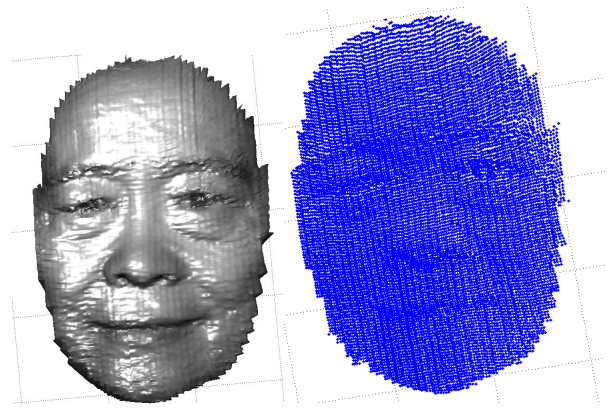


Fig. 7. The blue dots on the right side image are pixels in their real three dimensional space. The image on the left side has a light source to assist visualization the 3D reconstruction result.

In figure 7, a light source is added to the left side image to assist visualizing of the final result. This step helps us to visually verify our methodology. We also added the (R, G, B) value of the each pixel to the final reconstruction as shown in figure 8. The rise and fall of the human face can be clearly identified in figure 8. The current reconstruction only represent human faces. This work can be further completed by adding more 2D images that capture the entire head or human body. This is one of our possible future works.

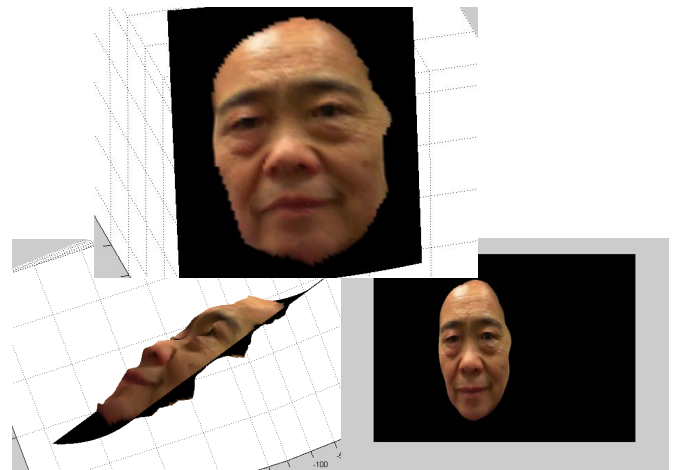


Fig. 8. Adding color (R, G, B) to each pixel and the final result.

IV. FUTURE WORK

A challenge for object reconstruction lays on precise facial expression reconstruction. Human face is a dynamic structure that changes its shape non-rigidly since muscles deform soft tissue and move bones. We plan to add more datasets and more images of human faces to test the algorithm.

Common improvement to the work includes but not limited to:

- Micro motion such as facial expressions. The proposed method can reconstruct a 3D object regardless of the facial expression as indicated in the previous section.

Preliminary testing and mathematic models are only based on stationary objects. This work focuses on 3D object reconstruction. Object recognition research using automatic or semi-automatic techniques received significant attention in the last two decades. Recent work shows vision-based motion capture can be detected and recognized but with low accuracy [13].

One important reason for the growing interest in 3D reconstruction is the wide range of possible applications for object recognition systems. Our plan to advance to object recognition can also be significantly affected by micro motion, such as the facial expressions on the appearance of the human or animal face. Human face is a dynamic structure that changes its shape non-rigidly since muscles deform soft tissue and move bones. Neurophysiologic studies have suggested that facial expression recognition happens in parallel to face identification [14]. Moving forward, the second stage of my work will focus on 3D face recognition.

- Reconstructing 3D objects data in different shapes and sizes. We have not tested overly large object which could be out of the 3D space we defined in section 2. Several existing literatures discusses the 3D construction of a historic artifacts using multiple (20-30) images. Existing software (such as Agisoft, <https://www.agisoft.com/>) designed for cultural heritage documentation or just visualization effect.
- Another possible challenge for object recognition is the distance/angle from the view point to the object. The existing camara and the absolute 3D coordinates are pre-calculated. To change the viewpoint of the camara, it requires recalibration, which could be a more efficient process. Recent studies have shown that recognition rates for unfamiliar objects drop significantly when there are different viewpoints for the training and test set [13]. However, there has been debate about whether object recognition is viewpoint-dependent or not [14]. My study will be using different objects with different viewpoints to test and verify this understanding.

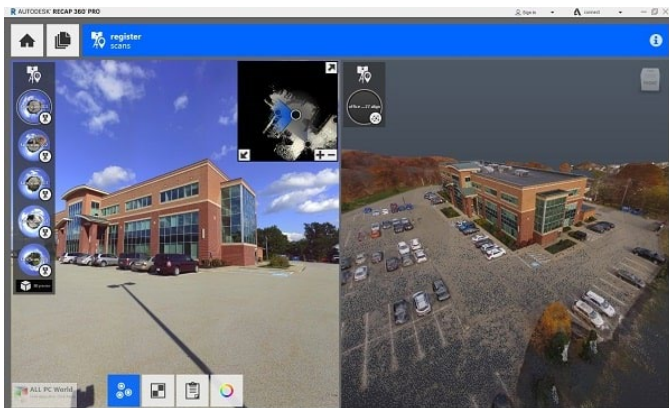


Fig. 9. Terrain modeling for example, where resolution is not the key (AutoDesk Pro)

- Addressing 3D objects occlusion, or covered with partial data hidden (cover with another object) e.g. part of the

object maybe occluded and not visible when images are taken from certain angles or because the subject grew a beard, is wearing glasses or a hat. Some existing techniques use deep learning methods to increase the final reconstruction resolution or possibly even add more details [15]. Our existing work focused on the precision reconstructions using existing data. We plan to address this challenge at a later stage, in particular, the threshold of occlusion and the effect on recognizing similar but different objects.

- Rendering speed is another problem. Current rendering of a human face takes roughly 5 minutes. This is because we want to precisely reconstruct the 3D object with every pixel's information preserved. Image interpolation is another possibility to increase the speed, but interpolation will lose image details [16].

Upon completion, this work will advance the understanding of reconstructing and recognizing 3D objects in a robust environment with limited lighting and camera.

V. CONCLUSION

This work discussed preliminary research in 3D human face reconstruction using two 2D images from two camaras. The work differs from previous research by adding precise reconstructions instead of interpolation, which does not represent the real data. In the near future, we plan to assess the possibilities of precise 3D object recognition with the provided methodology.

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